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# TME–305

## B. TECH. (MECHANICAL ENGINEERING)
## (THIRD SEMESTER)
## END SEMESTER EXAMINATION, Dec., 2023

MANUFACTURING PROCESSES–I

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain in brief pressure die casting with the help of neat sketch. (CO1 & CO2)

(b) The length of the down sprue leading into the runner of a mould = 200 mm. (CO1 & CO2)

The cross-sectional area at its base = 400 mm$^2$. Volume of the mould cavity = 0.0012 m$^3$. Determine :

(i) velocity of the molten metal flowing through the base of the down sprue

(ii) volume rate of flow

(iii) time required to fill the mould cavity

(c) When casting low carbon steel under certain mould conditions, the mould constant in Chvorinov's rule = 4.0 min/cm$^2$. Determine how long solidification will take for a rectangular casting whose length = 30 cm, width = 15 cm, and thickness = 20 mm. (CO1 & CO2)

2. (a) Define rolling of sheet with the help of diagram. Derive the relation for power input in rolling process. (CO2 & CO3)

(b) Wire is drawn through a draw die with entrance angle = 10°. Starting diameter is 4 mm and final diameter = 3 mm. The coefficient of friction at the work die interface = 0.07. The metal has a strength coefficient $k$ = 215 MPa and a strain hardening exponent $n$ = 0.25. Determine the draw stress and draw force in this operation. (CO2 & CO3)

(c) Explain the process of Extrusion. State the advantage of suing indirect extrusion over direct extrusion. (CO2 & CO3)

3. (a) Briefly explain the following terminology : (CO3 & CO4)

(i) Strain hardening

(ii) Back up roll

(iii) Flow stress

(iv) Yield strength of material

(b) Give suitable justification to differentiate the indirect extrusion and wire drawing. What the wire drawing is carried out in number of passes ? Justify. (CO3 & CO4)

(c) For the following given data, find the diameter of punch and die. Also find the blanking force. (CO3 & CO4)

Disc of diameter = 150 mm is cut from a sheet of thickness 3.2 mm. Shear strength of the sheet = 310 MPa

4. (a) Discuss the importance of low stress in forming process. Use relevant diagram to explain. (CO3 & CO4)

   (b) Discuss the process of extrusion with neat sketch. (CO3 & CO4)

   (c) State the importance of jigs and fixture in flexible manufacturing system. Elaborate. (CO3 & CO4)

5. (a) What is compaction and sintering in power metallurgy. Explain in detail. (CO4 & CO5)

   (b) Explain in detail the rapid sintering process. Suggest some applications where rapid sintering is applied. (CO4 & CO5)

   (c) What is conditioning monitoring process in power metallurgy ? Explain in brief. (CO4 & CO5)

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# TME–308

## B. TECH. (ME) (THIRD SEMESTER)

## END SEMESTER EXAMINATION, Dec., 2023

MANAGEMENT OF INTELLECTUAL PROPERTY RIGHTS

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define Intellectual Property (IP) and elaborate on its importance in fostering innovation and protecting creation. (CO3 & CO4)

(b) Explain the difference between industrial property and copyright, highlighting their distinctive features and areas of application. (CO3 & CO4)

(c) How do intellectual property rights serve as incentives for innovation and creativity in a knowledge-based economy ? (CO3 & CO4)

*P. T. O.*

2. (a) Differentiate between provisional and non-provisional patents and discuss the advantages and disadvantages of each. (CO2 & CO5)

   (b) What are the benefits of a patent portfolio ? List down the contents of a patent portfolio. (CO2 & CO5)

   (c) Discuss the essential components of an effective intellectual property management strategy for a tech startup and its role in sustaining business growth. (CO2 & CO5)

3. (a) Describe the process of patent prosecution from filling to grant, emphasizing the key stages involved and the challenges the applicants might face. (CO5 & CO4)

   (b) Define 'inventive step' under the Indian Patent Act. List down the various types of Patents. (CO5 & CO4)

   (c) Discuss the significance of patent examination in determining patentability and the role of patent examiners in this process. (CO5 & CO4)

4. (a) How do design rights differ from patents and trademarks ? Explain the scope of protection and the application of design rights in various industries. (CO4 & CO5)

   (b) Discuss the significance of industrial designs in product differentiation and consumer choice in the market. (CO4 & CO5)

   (c) Discuss in brief the historical evolution of IPR laws under International Regime. (CO4 & CO5)

5. (a) Explain the distinctiveness requirement for trademark registration. Provide examples of marks that might face challenges in meeting this requirement. (CO1 & CO2)

   (b) Compare and contrast geographical indications with trademarks, focusing on their unique protection and the impact on consumers' perception of product quality and origin. (CO1 & CO2)

   (c) Distinguish between Copyright and Industrial Design. (CO1 & CO2)



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# TME–306

## B. TECH. (ME) (THIRD SEMESTER)
## END SEMESTER EXAMINATION, Dec., 2023
### ENGINEERING MECHANICS

**Time : Three Hours**

**Maximum Marks : 100**

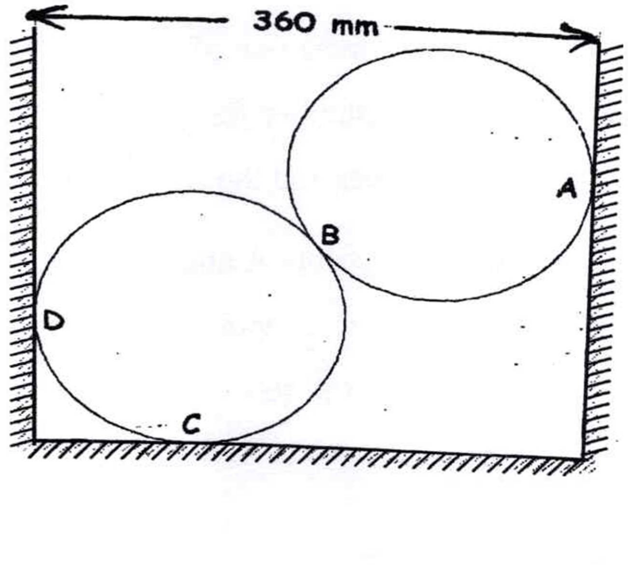
**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

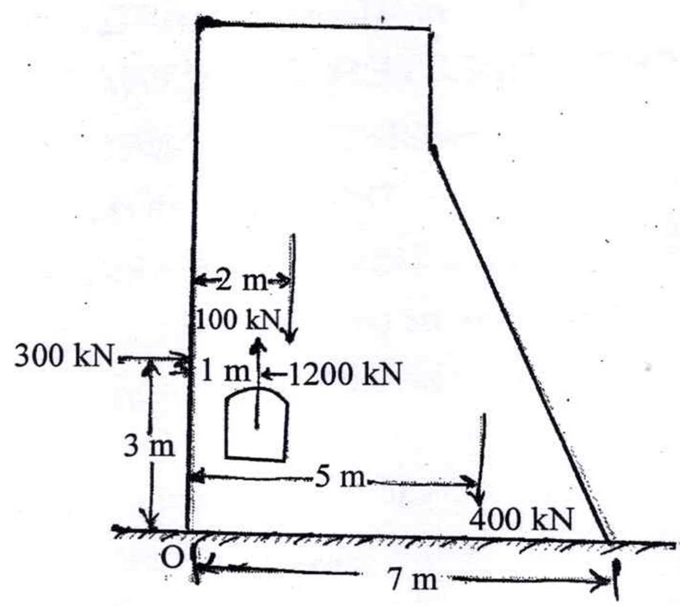
1. (a) Two smooth spheres each of radius 100 mm and weighing 100 N, rest in a horizontal channel having vertical walls, the distance between which is 360 mm. Find the reactions at the points of contact A, B, C and D as shown in Figure. (CO1 & CO2)




*P. T. O.*

(b) Calculate system of forces and give an example of each. (CO1 & CO2)

(c) Various forces to be considered for the stability analysis of a dam are shown in Figure. The dam is safe if the resultant force passes through middle third of the base. Verify whether the dam is safe. (CO1 & CO2)

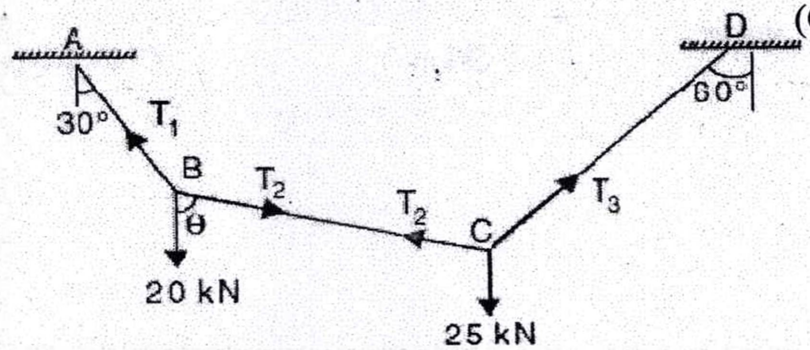



2. (a) Sate and explain law of transmissibility and parallelogram law of forces. (CO1 & CO2)

(b) The resultant of two forces one of which is 3 times the other is 300 N. when the direction of smaller force is reversed, the resultant is 200 N. Determine the two forces and the angle between them. (CO1 & CO2)

(c) A wire is fixed at two points A and D as shown in Figure. Two weights 20 kN and 25 kN are supported at B and C respectively. When equilibrium is reached it is found that inclination of AB is 30° and that of CD is 60° to the vertical. Determine the tension in the segments AB, BC and CD of the rope and also the inclination of BC to the vertical.

(CO1 & CO2)

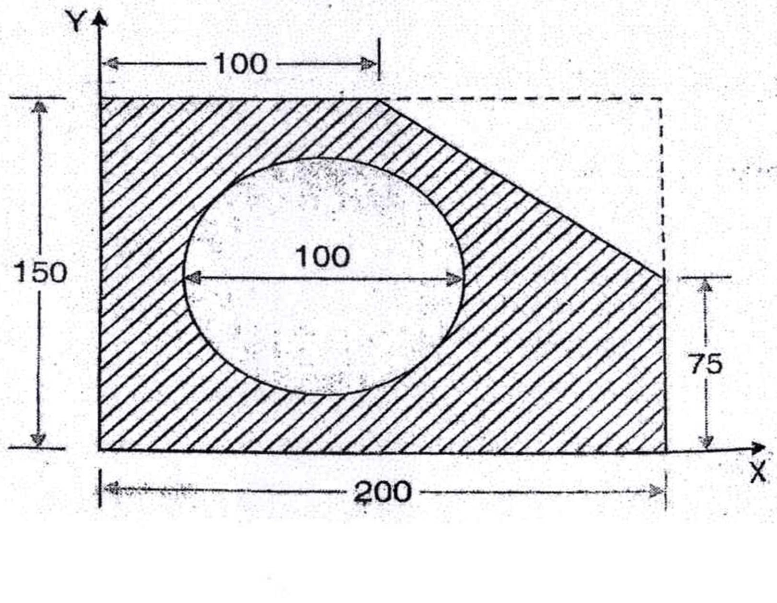



3. (a) Derive the relation for moment of inertia for the following cased :(CO3)

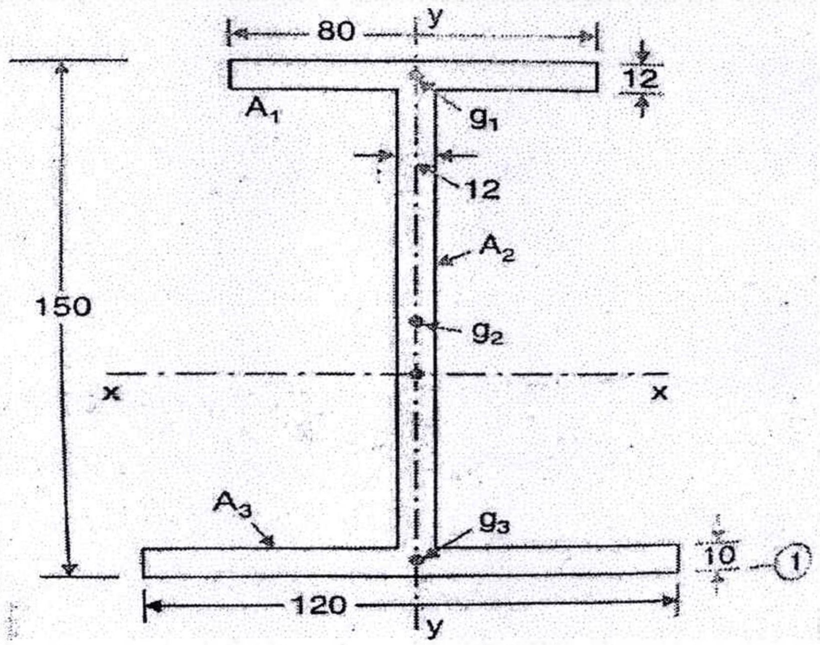
   (i) Moment of inertia of a triangle about its centroidal axis parallel to base.

   (ii) Movement of inertia of a circle about its diametral axis.

   (b) Determine the coordinates x and y of the centre of a 100 mm diameter of circular hole cut in a thin plate so that this point will be the centroid of the remaining shaded areas shown in Figure. (All dimensions are in mm). (CO3)

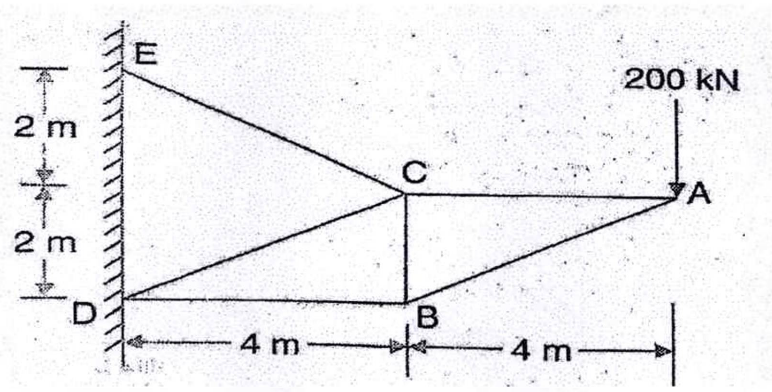


(c) Determine the polar moment of inertia about centroidal axis of the I-section shown in the figure. Also determine the radii of gyration with respect to x-x and y-y axis. (All dimensions are in mm). (CO3)

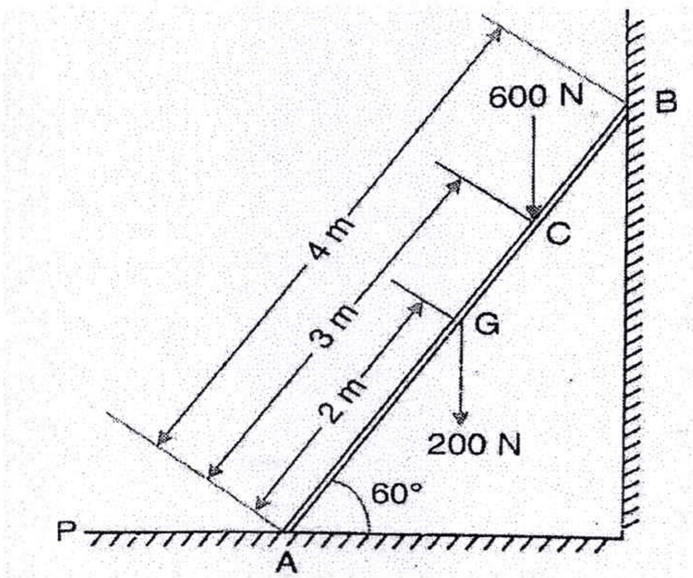



4. (a) Explain the procedure of method of joint and method of section for analysis of pin connected plane frame. (CO4 & CO5)

(b) Determine the forces in all members of the frame shown in figure. Indicate the nature of forces also. (CO4 & CO5)



(c) A ladder of length 4m, weighing 200N is placed against a vertical wall as shown in figure. The coefficient of friction between the wall and ladder is 0.2 and that between the floor and the ladder is 0.3. In addition to self weight, the ladder has to support a man weighing 600N at a distance of 3m from A. Calculate the minimum horizontal force to be applied at A to prevent slipping. (CO4 & CO5)




5. (a) Explain the following terms : (CO5 & CO6)

   (i) Sliding and rolling friction

   (ii) Laws of friction

   (iii) Angle of friction

   (iv) Angle of repose

   (v) Cone of friction

   (b) (i) Explain the Law of Machine. (CO5 & CO6)

   (ii) Discuss a reversible machine has efficiency greater than 50%.

(c) In a simple machine, whose velocity ratio is 30, a load of 2400N is lifted by an effort of 150N and a load of 3000N is lifted by an effort of 180N. Find the law of machine and calculate the load that could be lifted by a force of 200N. Calculate also : (CO5 & CO6)

(i) The amount of effort wasted in overcoming the friction

(ii) Mechanical Advantage

(iii) The efficiency

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# TME–304

## B. TECH. (ME) (THIRD SEMESTER)

## END SEMESTER EXAMINATION, Dec., 2023

BASIC THERMODYNAMICS

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define the following terms : (CO1)

(i) Thermodynamics

(ii) Homogeneous and heterogeneous system

(iii) Point/state functions and path function

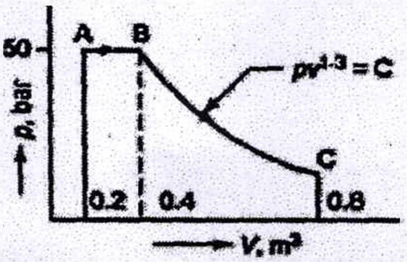
(iv) Intensive and extensive properties

(v) Reversible and irreversible process

(b) Derive an equation of work done and heat transfer for an isothermal and adiabatic process. (CO1, CO2)

*P. T. O.*

(c) Determine the total work done by a gas system following an expansion process as shown in Figure. (CO1, CO2)




2. (a) Explain the concept of PMM-I with the help of first law of thermodynamics. Also, prove that heat transfer is a path function and energy is a property. (CO1, CO2)

(b) A cylinder contains $0.12\ m^3$ of air at 1 bar and 90°C. It is compressed to $0.03\ m^3$. The final pressure being 6 bar. Find the index of compression, increase in internal energy and heat transfer. Take, $R = 0.287$ kJ/kg-K, & $C_v = 0.717$ kJ/kg-K. (CO2)

(c) Explain the following terms : (CO1, CO2, CO4)

(i) Pure substances

(ii) Steady flow energy equation

(iii) Throttling process

(iv) Work and heat transfer

(v) Enthalpy and its unit

3. (a) State the importance of the second law of thermodynamics. Also, explain the two statements of this law and with the help of these statements explain the concepts of PMM-II, heat engine and refrigerator. (CO3)

(b) A Carnot heat engine getting heat at 800 K is used to drive a Carnot refrigerator maintaining 280 K temperature. Both engine and refrigerator reject heat at some temp, T, when the heat given to engine is equal to the heat absorbed by the refrigerator. Determine the efficiency of engine and the COP of refrigerator. (CO3)

(c) (i) What do you mean by coefficient of performance ? (CO3, CO4)

Show that $(\text{COP})_{\text{HP}} = (\text{COP})_{\text{Ref}} + 1$

(ii) Explain the Carnot cycle with the help of T-s diagram. This cycle is an idea cycle which is practically not possible but still this cycle is being studied in thermodynamics, why ?

4. (a) What do you mean by Clausius inequality ? State and prove. (CO4)

(b) What is the concept of entropy ? Explain. Also, prove that $dS = \delta Q/T$ for a reversible process and discuss the principle of increase of entropy. (CO4)

(c) A block of iron weighing 100 kg and having a temperature of 100°C is immersed in 50 kg of water at a temperature of 20°C. What will be the change of entropy of the combined systm of iron and water ? Specific heats of iron and water are 0.45 and 4.18 kJ/kg-K respectively. (CO4)

5. (a) Explain the following terms : (CO1, CO4, CO5)

(i) Dalton's law and Amagat's law

(ii) Joule-Thomson coefficient.

(iii) Available and unavailable energy

(iv) Dead state and useful work

(v) Critical point and triple point

(b) What do you mean by thermodynamic relations ? Explain why are they important in thermodynamic and derive the Maxwell relations. (CO5)

(c) Calculate volume, density, enthalpy, and entropy of 2 kg of steam at 80°C and having a dryness faction of 0.85. (CO1)

**Properties of water :** At 80°C;

$P_{sat} = 47.39$ kPa $\quad v_f = 0.001029\ m^3/kg$

$v_g = 3.40715\ m^3/kg \quad h_f = 334.88$ kJ/kg

$h_{fg} = 2308.77$ kJ/kg-K $\quad s_f = 1.0752$ kJ/kg-K

$s_{fg} = 6.5369$ kJ/kg-K

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# TMA–303

## B. TECH. MECHANICAL ENGINEERING
## (THIRD SEMESTER)
## END SEMESTER EXAMINATION, Dec., 2023

ENGINEERING MATHEMATICS-III

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Is the function : (CO1)

$$u(x, y) = \frac{1}{2}\log(x^2 + y^2)$$

harmonic ? If yes, then find its harmonic conjugate function.

(b) Check the analyticity of the function : (CO1)

$$f(z) = z|z|$$

(c) Using Newton-Raphson method to compute a root of the equation :

$$x^3 - x - 1 = 0$$

with initial condition $x_0 = 1.6$ (Correct to three decimal places).

(CO3)

*P. T. O.*

2. (a) Use Cauchy's integral formula to evaluate :

$$\int_C \frac{\cos \pi z + \sin \pi z}{(z-1)(z-2)} dz$$

where $c$ is the circle $|z| = 4$. (CO2)

(b) Evaluate the complex integral :

$$\int_{1-i}^{2+3i} (z + z^2)\, dz$$

along the line joining the points (1, – 1) and (2, 3). (CO2)

(c) Given that the initial value problem : (CO5)

$$\frac{dy}{dx} = x - y$$

with initial condition $y(0) = 1$. Use Runge-Kutta method to obtain y for $x = 0.2$ correct to three decimal places (Taking step size $h = 0.2$).

3. (a) Use Bisection method, to find a real root of an equation :

$$x \log_{10} x = 1.2$$

correct to three decimals. (CO3)

(b) A train is moving at the speed of 30 m/sec. Suddenly brakes are applied. The speed of the train per second after '$t$' seconds is given below :

| Time ($t$) | Speed ($v$) |
|---|---|
| 0 | 30 |
| 5 | 24 |
| 10 | 19 |
| 15 | 16 |
| 20 | 13 |
| 25 | 11 |
| 30 | 10 |
| 35 | 8 |
| 40 | 7 |
| 45 | 5 |

Use Simpson's $\frac{\text{'3'}}{8}$ rule to complete the distance moved by the train in 45 seconds. (CO3)

(c) Define absolute, relative and percentage errors. Find the relative error if $\frac{2}{3}$ is approximated to 0.667. (CO3)

4. (a) Solve :

$$\frac{dy}{dx} = xy$$

numerically from $x = 0$ to $x = 1$ using Euler method with initial condition of $y(0) = 1$ and step size ($h$) of 0.2. Find the value of $y$ (1) (Correct to three decimal places). (CO5)

(b) A curve is drawn to pass through the points given by the following table :

| $x$ | $f(x)$ |
|---|---|
| 1 | 2 |
| 1.5 | 2.4 |
| 2.0 | 2.7 |
| 2.5 | 2.8 |
| 3.0 | 3 |
| 3.5 | 2.6 |
| 4.0 | 2.1 |

Find the value of the integral $\int_1^4 f(x)\,dx$ using Simpson's rule.

(c) Consider an initial value problem :

$$\frac{du}{dt} = 3t^2 + 1;\ u(0) = 0$$

Solve the given problem by numerically using Euler's method, calculate $u(0.2)$ taking step-size ($h = 0.1$) (CO5)

5. (a) If the probability density function of a continuous random variable X is given by : (CO6)

$$f(x)=\begin{cases} kx^2; & 0<x<1 \\ 0; & \text{otherwise} \end{cases}$$

find the value of $k$ and hence compute its mean, variance, and standard deviation.

(b) Following are the data on the drying time of a certain varnish and amount of an additive that is intended to reduce the drying time :

| **Amount of varnish additive groups ($x$)** | **Drying time ($y$ in yours)** |
|---|---|
| 0 | 12.0 |
| 1 | 10.5 |
| 2 | 10.0 |
| 3 | 8.0 |
| 4 | 7.0 |
| 5 | 8.0 |
| 6 | 7.5 |
| 7 | 8.5 |
| 8 | 9.5 |

(i) Fit a simple linear regression line of $y$ on $x$.

(ii) Use the result of part (i) to predict the drying time of the varnish when 6.5 gm of additive is being used.

(c) The following are the values of two variables, diastolic blood pressure ($y$) and time in minutes ($x$). Diastolic blood pressure ($y$) was recorded at various time periods (x) on individuals. Find the regression line of $y$ on $x$ and $x$ on $y$.

| **Time periods ($x$)** | **Diastolic blood pressure ($y$)** |
|---|---|
| 0 | 72 |
| 5 | 67 |
| 10 | 70 |
| 15 | 65 |
| 20 | 66 |



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# TME–302

## B. TECH. (ME) (THIRD SEMESTER)

## END SEMESTER EXAMINATION, Dec., 2023

MATERIALS SCIENCE

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is the importance of Miller indices ? Detail out the procedure to obtain Miller-Bravais in a hexagonal crystal. (CO1)

   (b) Derive the expression for atomic packing factor for a Simple Cubic, Body centered cubic, and face centered cubic structure. (CO1)

   (c) Determine the interplanar spacing between (200), (220) and (111) planes in a FCC crystal. The atomic radius is 1.246 Å. (CO1)

2. (a) Explain the different types of defects and explain the surface defects with the help of neat sketches. (CO2)

*P. T. O.*

(b) Draw Stress-Strain Diagram for Mild steel and explain all the important points on it. (CO2)

(c) Draw the miller indices for the following planes : (CO2)

$(3\ 3\ 1)\ (1\ 0\ 1)\ (\bar{2}\ 3\ 1)\ (0\ 0\ 2)\ (1\ \bar{1}\ 1)$

3. (a) Draw the iron carbon equilibrium diagram and show their salient features. Indicate significance of this diagram for heat treatment of steel. (CO3, CO4)

(b) Explain and draw a typical creep curve of stress versus time at constant stress and constant elevated temperature clearly showing the instantaneous deformation, three stages of creep and rupture. At what temperature does creep become significant ? (CO3, CO4)

(c) (i) List properties required for the material to withstand high temperature. Discuss the use of alloy steels as heat resistant materials.

(ii) Explain ceramics materials, their characteristics, basic structure, and applications. (CO3, CO4)

4. (a) Sketch and explain the TTT diagram for Eutectoid steel. Depict important transformations taking place in it during cooling. (CO5)

(b) Describe the following heat treatment procedure : (CO5)

(i) Full annealing

(ii) Flame hardening

(iii) Nitriding

(iv) Tempering

(v) Cyaniding

(c) Explain the structure of Austenite, Pearlite, Martensite, Bainite with a neat sketch. (CO5)

5. (a) Explain the corrosion mechanism. What are the different methods of preventing corrosion ? Explain with the aid of sketches wherever necessary. (CO6)

(b) (i) List properties required for the material to withstand high temperature. Discuss the use of alloy steels as heat resistant materials

(ii) Explain ceramics materials, their characteristics, basic structure, and applications. (CO6)

(c) Bring out differences between the following terms : (CO6)

(i) Dry and Wet corrosion

(ii) Ductile fracture and brittle fracture

(iii) Eutectoid and eutectic reactions

(iv) Austempering and martempering

(v) Alpha iron and Delta iron

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# TMS–301

## B. TECH. (MECHANICAL ENGINEERING)

## (THIRD SEMESTER)

## END SEMESTER EXAMINATION, Dec., 2023

PRODUCT DESIGN AND DEVELOPMENT

**Time : Three Hours**

**Maximum Marks : 100**

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the concept of 'Product Definition'. Also discuss the benefits of multifunctional products. (CO1, CO2)

   (b) Illustrate in detail the Stage Gate Process of Product Development. (CO1, CO2)

   (c) Describe in detail about the concept of 'Product Specification'. Also discuss different steps necessary while defining product specification. (CO1, CO2)

*P. T. O.*

2. (a) Discuss in detail about the concept of 'Product Architecture'. (CO2, CO3)

   (b) Explain in detail about the concept of 'Teardown and Benchmarking'. (CO2, CO3)

   (c) Describe in detail about the concept of 'Advanced Manufacturing'. (CO2, CO3)

3. (a) Discuss in detail about the concept of 'Idea Generation'. (CO3, CO4)

   (b) Describe in detail about the concept of 'Creativity and Problem Solving'. (CO3, CO4)

   (c) Explain in detail about the concept of 'Conceptual Decomposition' with suitable benefits. (CO3, CO4)

4. (a) Discuss in detail about the concept of 'Industrial Design' and 'Perceived Quality'. (CO4, CO5)

   (b) Explain in detail about the concept of 'Ergonomically considered design' alongwith suitable benefits. (CO4, CO5)

   (c) Describe in detail about the concept of 'Detailed design'. (CO4, CO5)

5. (a) Discuss in detail about the concept of 'Engineering Bill of Materials'. (CO5, CO6)

   (b) Describe in detail about the concept of Process Failure Mode and Effects Analysis (PFMEA). (CO5, CO6)

   (c) Explain in detail about the concept of Product Life-cycle Management. Also discuss in brief about the concept of 'Product upgrades'. (CO5, CO6)

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# TAE–301

## B. TECH. (THIRD SEMESTER)

## END SEMESTER EXAMINATION, Dec., 2023

AUTOMOTIVE SYSTEMS

**Time : Three Hours**

**Maximum Marks : 100**

**Note :** (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the evolution of automobiles from their inception to the present day, highlighting key technological advancements and their impact on society. (CO4, CO3 & CO4)

   (b) Enumerate and elaborate on the diverse applications of automobiles in different sectors (e.g., transportation, agriculture, emergency services). How do these applications showcase the versatility of automobiles ? (CO4, CO3 & CO4)

   (c) How has the automotive industry's innovation in recent years influenced the applications of automobiles in areas such as smart cities, autonomous vehicles, and sustainability initiatives ?(CO4, CO3 & CO4)

*P. T. O.*

2. (a) Describe the major components that form the aggregates of an automobile. How do these aggregates work together to ensure the vehicle's functionality and performance ? (CO2, CO4, CO5 & CO6)

   (b) Compare and contrast the components of manual and automatic transmissions, highlighting their distinctive mechanisms and applications in different types of vehicles. (CO2, CO4, CO5 & CO6)

   (c) Explain the importance of suspension systems in vehicles and the role of their various components in ensuring a comfortable and safe ride.

   (CO2, CO4, CO5 & CO6)

3. (a) What are the significant factors influencing automotive architecture ? Discuss how these factors impact vehicle design, safety, and performance. (CO3, CO5 & CO6)

   (b) Analyze the latest trends in the automotive industry, such as electric vehicles, autonomous driving, and connectivity features. How are these trends shaping the future of automobiles ? (CO3, CO5 & CO6)

   (c) Highlight the key subsystems of an automobile and their individual importance in ensuring the vehicle's overall performance and safety.

   (CO3, CO5 & CO6)

4. (a) Compare and contrast electric vehicle transmissions with those found in conventional automobiles, emphasizing the key differences.

   (CO1, CO2 & CO5)

   (b) Discuss the role of the transmission system in an automobile and how different transmission types impact the vehicle's performance and efficiency. (CO1, CO2 & CO5)

(c) How does the chassis contribute to the structural integrity and safety of a vehicle, and what are the crucial components involved in the chassis system ? (CO1, CO2 & CO5)

5. (a) Define benchmarking and elaborate on its significance in the automotive industry. (CO1, CO2, CO5 & CO6)

(b) Analyze the career opportunities available in the field of benchmarking within the automotive industry. What skill sets are essential for a successful career in this domain ? (CO1, CO2, CO5 & CO6)

(c) Explain how the practice of teardown and benchmarking expedites the release of new automobiles, emphasizing its impact on the development process. (CO1, CO2, CO5 & CO6)

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